

Luis Garcia Rivera

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EDUCATION

Washington University in St. Louis, McKelvey School of Engineering
B.S. Mechanical Engineering; Minor in Robotics & Mechatronics
GPA: 3.79/4.00 | Dean's List | HSF Scholar | Kessler Scholar

St. Louis, MO
Expected Dec 2027

ENGINEERING EXPERIENCE

Manufacturing Equipment Engineer Co-Op – Tesla

Fremont, CA

Model 3 Body in White Manufacturing

Jan 2026 – Aug 2026

- Redesigned bowl-feeder end-of-arm-tool locating hardware after new feeder geometry blocked a critical locator; created **2 variants for 3 fixtures** using NAAMS and custom 17-4 PH components to restore full **6-DOF part constraint**; carried the design through machining, finishing, assembly, and installation, passed a **25-part live-line validation with no issues**, and remained stable through one week of production monitoring as related pick faults trended down.
- Diagnosed a separate clamp-misalignment fault that allowed parts to slide past a locating pin; redesigned the clamp with a weld-bead clearance chamfer, machined it in-house, and installed it, substantially reducing the recurring production fault.
- Partnered with two senior engineers to cut station cycle time from **75s to 65s (13%)** in a three-robot FANUC spot-weld sequence; reprogrammed an adjacent robot in teach mode to assume one weld and its approach/home points from the bottleneck robot, then validated interference, full-speed operation, and production stability with maintenance.
- Reverse-engineered an OEM three-clamp fixture system used at **8 production locations** into internal CATIA V5 models and GD&T drawings, creating an in-house fabrication path versus an approximately **\$88K replacement package** and quoted **10-month lead time**.

Optimus Training Data Collection – Modular Factory Hardware

May 2026 – Aug 2026

- Co-developed and deployed modular training-data hardware to **20 mounted Fremont stations**; co-led surveys, assembly, wiring, commissioning, troubleshooting, and camera-angle iteration around live production constraints.
- Owned CATIA V5 mechanical design for **3 adjustable camera-mount architectures** and tool-free, 3D-printed compute-module and battery-pack mounts with locating pins and swappable interfaces for common factory structures; released designs to a print farm and standardized hardware for repeatable deployment and handoff.

Humanoid Robotics Training Facility – Layout & Handoff Systems

Mar 2026 – Aug 2026

- Owned the layout, equipment-tracking, and demolition-documentation workstream for a planned **40-station** training facility; produced **21 contractor-ready station layouts** with **179 equipment callouts** and coordinated execution with architects, facilities, controls engineers, and contractors; integrated conveyor, utility, and controls-adjacent requirements into the permitted layout.

Undergraduate Researcher – Computational Mechanics & AI Lab

St. Louis, MO

Applied Machine Learning in Fluid Dynamics

May 2025 – Aug 2025

- Developed physics-informed neural networks in PyTorch for cardiovascular flow prediction, achieving up to a **1000× reduction in prediction time**; modernized a legacy Fortran solver into a modular Python pipeline for reproducible validation and rapid parameter studies.

Structures Engineer – WashU Design/Build/Fly (AIAA)

St. Louis, MO

Composite Airframe Design and Test

Aug 2024 – Present

- Designed load-bearing composite airframe structures in SolidWorks and used ANSYS finite-element analysis to validate safety factors while minimizing takeoff weight; led fabrication and destructive testing to correlate physical failures with simulation predictions.

TECHNICAL SKILLS

Mechanical Design & Manufacturing: CATIA V5/3DEXPERIENCE, SolidWorks (CSWA), AutoCAD/DWG, GD&T, DFM/DFA, fixtures and end-of-arm tooling, 3D printing, mill/lathe machining, composite fabrication

Automation & Test: FANUC/KUKA robot programming and teach-pendant operation, Allen-Bradley GuardLogix PLC, SCADA, pneumatics, sensor integration, commissioning and production validation

Analysis & Software: ANSYS FEA/CFD, Python (PyTorch, NumPy, Pandas, Matplotlib), C++/Arduino, MATLAB, root-cause analysis, DOE, OEE optimization

PROJECT

Autonomous Mobile Robot – Embedded Controls Project

Sep 2024 – Dec 2024

- Designed and prototyped an Arduino/C++ autonomous vehicle integrating sensors, a DC-motor drivetrain, and PID control; achieved **95% path-following accuracy** in autonomous operation.

LEADERSHIP & ACTIVITIES

Teaching Fellow: Engineering Circuits | **Tutor:** WashU Learning Center | **Undergraduate Advisory Board**